

4 Lysosomes are membrane-bound organelles found in mammalian cells.

- (a) Scientists measured the concentration of cholesterol in the membranes of lysosomes in a mammalian cell.

The concentration of cholesterol in the lysosome membranes was found to be lower than the concentration in other membranes inside mammalian cells.

- (i) State how a lower concentration of cholesterol would make the properties of lysosome membranes different from other membranes in mammalian cells.

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..... [1]

- (ii) Suggest why the lower concentration of cholesterol in lysosome membranes would help lysosomes carry out their function.

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..... [1]

- (b) The enzyme α -galactosidase is present in lysosomes.

Students investigated the effect of substrate concentration on the rate of reaction catalysed by α -galactosidase at pH 4.5 and at pH 5.9.

The results are shown in Fig. 4.1.

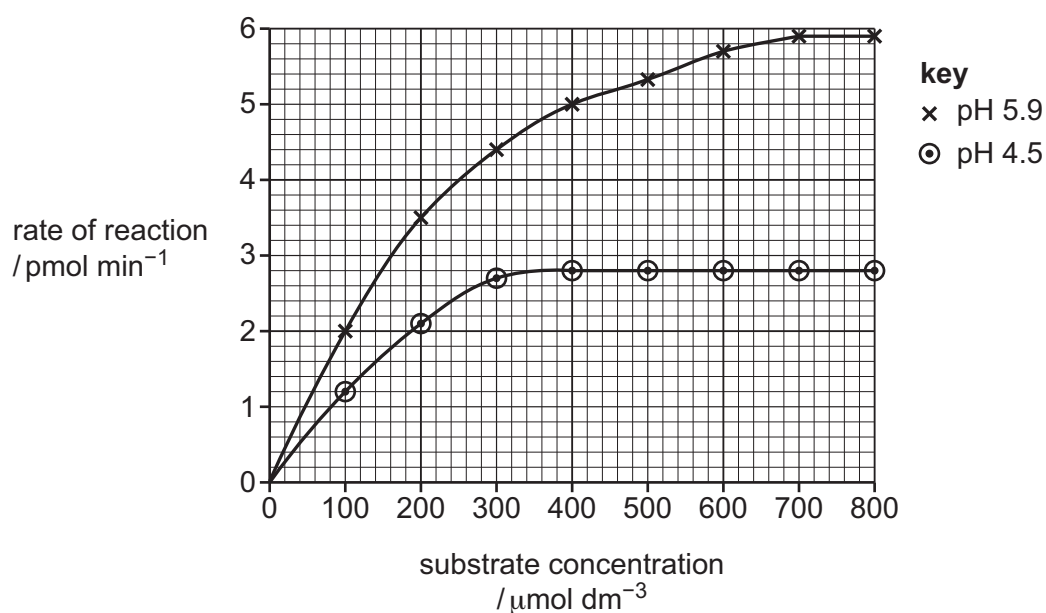


Fig. 4.1



- (i) Determine the Michaelis–Menten constant, K_m , for α -galactosidase at pH 4.5 using the data in Fig. 4.1.

State the unit for the K_m value in your answer.

K_m = unit [2]

- (ii) With reference to Fig. 4.1, describe the differences in the results at pH 4.5 and pH 5.9 **and** suggest explanations for the differences.

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[Total: 8]